
Neural Discrete Representation Learning

Aaron van den Oord
DeepMind
avdnoord@google.com

Oriol Vinyals
DeepMind
vinyals@google.com

Koray Kavukcuoglu
DeepMind
korayk@google.com

Abstract

Learning useful representations without supervision remains a key challenge in machine learning. In this paper, we propose a simple yet powerful generative model that learns such discrete representations. Our model, the Vector Quantised-Variational AutoEncoder (VQ-VAE), differs from VAEs in two key ways: the encoder network outputs discrete, rather than continuous, codes; and the prior is learnt rather than static. In order to learn a discrete latent representation, we incorporate ideas from vector quantisation (VQ). Using the VQ method allows the model to circumvent issues of “posterior collapse” — where the latents are ignored when they are paired with a powerful autoregressive decoder — typically observed in the VAE framework. Pairing these representations with an autoregressive prior, the model can generate high quality images, videos, and speech as well as doing high quality speaker conversion and unsupervised learning of phonemes, providing further evidence of the utility of the learnt representations.

1 Introduction

Recent advances in generative modelling of images [38, 12, 13, 22, 10], audio [37, 26] and videos [20, 11] have yielded impressive samples and applications [24, 18]. At the same time, challenging tasks such as few-shot learning [34], domain adaptation [17], or reinforcement learning [35] heavily rely on learnt representations from raw data, but the usefulness of generic representations trained in an unsupervised fashion is still far from being the dominant approach.

Maximum likelihood and reconstruction error are two common objectives used to train unsupervised models in the pixel domain, however their usefulness depends on the particular application the features are used in. Our goal is to achieve a model that conserves the important features of the data in its latent space while optimising for maximum likelihood. As the work in [7] suggests, the best generative models (as measured by log-likelihood) will be those without latents but a powerful decoder (such as PixelCNN). However, in this paper, we argue for learning discrete and useful latent variables, which we demonstrate on a variety of domains.

Learning representations with continuous features have been the focus of many previous work [16, 39, 6, 9] however we concentrate on discrete representations [27, 33, 8, 28] which are potentially a more natural fit for many of the modalities we are interested in. Language is inherently discrete, similarly speech is typically represented as a sequence of symbols. Images can often be described concisely by language [40]. Furthermore, discrete representations are a natural fit for complex reasoning, planning and predictive learning (e.g., if it rains, I will use an umbrella). While using discrete latent variables in deep learning has proven challenging, powerful autoregressive models have been developed for modelling distributions over discrete variables [37].

In our work, we introduce a new family of generative models successfully combining the variational autoencoder (VAE) framework with discrete latent representations through a novel parameterisation of the posterior distribution of (discrete) latents given an observation. Our model, which relies on vector quantization (VQ), is simple to train, does not suffer from large variance, and avoids the

“posterior collapse” issue which has been problematic with many VAE models that have a powerful decoder, often caused by latents being ignored. Additionally, it is the first discrete latent VAE model that get similar performance as its continuous counterparts, while offering the flexibility of discrete distributions. We term our model the VQ-VAE.

Since VQ-VAE can make effective use of the latent space, it can successfully model important features that usually span many dimensions in data space (for example objects span many pixels in images, phonemes in speech, the message in a text fragment, etc.) as opposed to focusing or spending capacity on noise and imperceptible details which are often local.

Lastly, once a good discrete latent structure of a modality is discovered by the VQ-VAE, we train a powerful prior over these discrete random variables, yielding interesting samples and useful applications. For instance, when trained on speech we discover the latent structure of language without any supervision or prior knowledge about phonemes or words. Furthermore, we can equip our decoder with the speaker identity, which allows for speaker conversion, i.e., transferring the voice from one speaker to another without changing the contents. We also show promising results on learning long term structure of environments for RL.

Our contributions can thus be summarised as:

- Introducing the VQ-VAE model, which is simple, uses discrete latents, does not suffer from “posterior collapse” and has no variance issues.
- We show that a discrete latent model (VQ-VAE) perform as well as its continuous model counterparts in log-likelihood.
- When paired with a powerful prior, our samples are coherent and high quality on a wide variety of applications such as speech and video generation.
- We show evidence of learning language through raw speech, without any supervision, and show applications of unsupervised speaker conversion.

2 Related Work

In this work we present a new way of training variational autoencoders [23, 32] with discrete latent variables [27]. Using discrete variables in deep learning has proven challenging, as suggested by the dominance of continuous latent variables in most of current work – even when the underlying modality is inherently discrete.

There exist many alternatives for training discrete VAEs. The NVIL [27] estimator use a single-sample objective to optimise the variational lower bound, and uses various variance-reduction techniques to speed up training. VIMCO [28] optimises a multi-sample objective [5], which speeds up convergence further by using multiple samples from the inference network.

Recently a few authors have suggested the use of a new continuous reparametrisation based on the so-called Concrete [25] or Gumbel-softmax [19] distribution, which is a continuous distribution and has a temperature constant that can be annealed during training to converge to a discrete distribution in the limit. In the beginning of training the variance of the gradients is low but biased, and towards the end of training the variance becomes high but unbiased.

None of the above methods, however, close the performance gap of VAEs with continuous latent variables where one can use the Gaussian reparameterisation trick which benefits from much lower variance in the gradients. Furthermore, most of these techniques are typically evaluated on relatively small datasets such as MNIST, and the dimensionality of the latent distributions is small (e.g., below 8). In our work, we use three complex image datasets (CIFAR10, ImageNet, and DeepMind Lab) and a raw speech dataset (VCTK).

Our work also extends the line of research where autoregressive distributions are used in the decoder of VAEs and/or in the prior [14]. This has been done for language modelling with LSTM decoders [4], and more recently with dilated convolutional decoders [42]. PixelCNNs [29, 38] are convolutional autoregressive models which have also been used as distribution in the decoder of VAEs [15, 7].

Finally, our approach also relates to work in image compression with neural networks. Theis et. al. [36] use scalar quantisation to compress activations for lossy image compression before arithmetic encoding. Other authors [1] propose a method for similar compression model with vector quantisation.

The authors propose a continuous relaxation of vector quantisation which is annealed over time to obtain a hard clustering. In their experiments they first train an autoencoder, afterwards vector quantisation is applied to the activations of the encoder, and finally the whole network is fine tuned using the soft-to-hard relaxation with a small learning rate. In our experiments we were unable to train using the soft-to-hard relaxation approach from scratch as the decoder was always able to invert the continuous relaxation during training, so that no actual quantisation took place.

3 VQ-VAE

Perhaps the work most related to our approach are VAEs. VAEs consist of the following parts: an encoder network which parameterises a posterior distribution $q(z|x)$ of discrete latent random variables z given the input data x , a prior distribution $p(z)$, and a decoder with a distribution $p(x|z)$ over input data.

Typically, the posteriors and priors in VAEs are assumed normally distributed with diagonal covariance, which allows for the Gaussian reparametrisation trick to be used [32, 23]. Extensions include autoregressive prior and posterior models [14], normalising flows [31, 10], and inverse autoregressive posteriors [22].

In this work we introduce the VQ-VAE where we use discrete latent variables with a new way of training, inspired by vector quantisation (VQ). The posterior and prior distributions are categorical, and the samples drawn from these distributions index an embedding table. These embeddings are then used as input into the decoder network.

3.1 Discrete Latent variables

We define a latent embedding space $e \in R^{K \times D}$ where K is the size of the discrete latent space (i.e., a K -way categorical), and D is the dimensionality of each latent embedding vector e_i . Note that there are K embedding vectors $e_i \in R^D$, $i \in 1, 2, \dots, K$. As shown in Figure 1, the model takes an input x , that is passed through an encoder producing output $z_e(x)$. The discrete latent variables z are then calculated by a nearest neighbour look-up using the shared embedding space e as shown in equation 1. The input to the decoder is the corresponding embedding vector e_k as given in equation 2. One can see this forward computation pipeline as a regular autoencoder with a particular non-linearity that maps the latents to 1-of- K embedding vectors. The complete set of parameters for the model are union of parameters of the encoder, decoder, and the embedding space e . For sake of simplicity we use a single random variable z to represent the discrete latent variables in this Section, however for speech, image and videos we actually extract a 1D, 2D and 3D latent feature spaces respectively.

The posterior categorical distribution $q(z|x)$ probabilities are defined as one-hot as follows:

$$q(z = k|x) = \begin{cases} 1 & \text{for } k = \operatorname{argmin}_j \|z_e(x) - e_j\|_2, \\ 0 & \text{otherwise} \end{cases}, \quad (1)$$

where $z_e(x)$ is the output of the encoder network. We view this model as a VAE in which we can bound $\log p(x)$ with the ELBO. Our proposal distribution $q(z = k|x)$ is deterministic, and by defining a simple uniform prior over z we obtain a KL divergence constant and equal to $\log K$.

The representation $z_e(x)$ is passed through the discretisation bottleneck followed by mapping onto the nearest element of embedding e as given in equations 1 and 2.

$$z_q(x) = e_k, \quad \text{where } k = \operatorname{argmin}_j \|z_e(x) - e_j\|_2 \quad (2)$$

3.2 Learning

Note that there is no real gradient defined for equation 2, however we approximate the gradient similar to the straight-through estimator [3] and just copy gradients from decoder input $z_q(x)$ to encoder output $z_e(x)$. One could also use the subgradient through the quantisation operation, but this simple estimator worked well for the initial experiments in this paper.